

Selected robustness metrics

Continuous values: mean +/- sample SD; control roughness is mean per transition (lower is smoother); roughness/effort/steps/runtime: successful trials only

Case / variant	trials	Success	Control roughness overall (mean/transition)	Control roughness after wall (mean/transition)	Control effort	Number of steps	Average time per step (s)
condition=dynamic_wall variant=control_bank_mppi_128	400	319/400 (79.8%)	0.281 +/- 0.064 (n=319)	0.229 +/- 0.080 (n=319)	260.885 +/- 29.211 (n=319)	51.978 +/- 7.456 (n=319)	0.005 +/- 0.001 (n=319)
condition=dynamic_wall variant=corridor_prior_map	400	290/400 (72.5%)	0.226 +/- 0.066 (n=290)	0.212 +/- 0.074 (n=290)	258.725 +/- 24.478 (n=290)	50.134 +/- 9.819 (n=290)	0.004 +/- 0.001 (n=290)
condition=dynamic_wall variant=frenet_corridor	400	285/400 (71.2%)	0.170 +/- 0.045 (n=285)	0.182 +/- 0.056 (n=285)	235.908 +/- 35.580 (n=285)	57.105 +/- 7.044 (n=285)	0.005 +/- 0.001 (n=285)
condition=dynamic_wall variant=full_swarm_prior_map	400	313/400 (78.2%)	0.211 +/- 0.068 (n=313)	0.222 +/- 0.081 (n=313)	242.945 +/- 34.962 (n=313)	57.105 +/- 8.642 (n=313)	0.006 +/- 0.001 (n=313)
condition=dynamic_wall variant=gaussian_prior_map	400	262/400 (65.5%)	0.159 +/- 0.046 (n=262)	0.175 +/- 0.058 (n=262)	231.063 +/- 36.898 (n=262)	57.615 +/- 7.164 (n=262)	0.006 +/- 0.001 (n=262)
condition=dynamic_wall variant=mode_selecting_corridor	400	298/400 (74.5%)	0.218 +/- 0.067 (n=298)	0.194 +/- 0.074 (n=298)	264.930 +/- 22.012 (n=298)	48.594 +/- 9.203 (n=298)	0.009 +/- 0.002 (n=298)
condition=dynamic_wall variant=mode_selecting_gaussian	400	273/400 (68.2%)	0.159 +/- 0.039 (n=273)	0.172 +/- 0.046 (n=273)	233.103 +/- 35.948 (n=273)	57.531 +/- 7.118 (n=273)	0.011 +/- 0.003 (n=273)
condition=dynamic_wall variant=standard_mppi_128	400	54/400 (13.5%)	0.165 +/- 0.027 (n=54)	0.146 +/- 0.028 (n=54)	257.152 +/- 5.275 (n=54)	38.370 +/- 0.958 (n=54)	0.002 +/- 0.000 (n=54)
condition=dynamic_wall variant=standard_mppi_128_runtime	400	90/400 (22.5%)	0.139 +/- 0.039 (n=90)	0.117 +/- 0.028 (n=90)	261.491 +/- 4.924 (n=90)	38.178 +/- 1.320 (n=90)	0.011 +/- 0.002 (n=90)
condition=no_wall variant=control_bank_mppi_128	100	100/100 (100.0%)	0.301 +/- 0.065 (n=100)	-	243.754 +/- 23.468 (n=100)	50.240 +/- 7.012 (n=100)	0.003 +/- 0.000 (n=100)
condition=no_wall variant=corridor_prior_map	100	98/100 (98.0%)	0.218 +/- 0.062 (n=98)	-	244.182 +/- 17.552 (n=98)	44.806 +/- 4.441 (n=98)	0.002 +/- 0.000 (n=98)
condition=no_wall variant=frenet_corridor	100	98/100 (98.0%)	0.167 +/- 0.042 (n=98)	-	214.378 +/- 26.316 (n=98)	55.490 +/- 7.076 (n=98)	0.003 +/- 0.000 (n=98)
condition=no_wall variant=full_swarm_prior_map	100	100/100 (100.0%)	0.215 +/- 0.063 (n=100)	-	222.475 +/- 27.439 (n=100)	53.760 +/- 5.852 (n=100)	0.002 +/- 0.002 (n=100)
condition=no_wall variant=gaussian_prior_map	100	99/100 (99.0%)	0.153 +/- 0.039 (n=99)	-	211.557 +/- 27.139 (n=99)	56.111 +/- 7.468 (n=99)	0.004 +/- 0.001 (n=99)
condition=no_wall variant=mode_selecting_corridor	100	100/100 (100.0%)	0.190 +/- 0.054 (n=100)	-	252.994 +/- 16.283 (n=100)	42.700 +/- 4.373 (n=100)	0.004 +/- 0.001 (n=100)
condition=no_wall variant=mode_selecting_gaussian	100	99/100 (99.0%)	0.153 +/- 0.034 (n=99)	-	212.318 +/- 26.415 (n=99)	56.030 +/- 6.947 (n=99)	0.006 +/- 0.001 (n=99)
condition=no_wall variant=standard_mppi_128	100	39/100 (39.0%)	0.161 +/- 0.028 (n=39)	-	256.095 +/- 5.249 (n=39)	38.179 +/- 0.970 (n=39)	0.001 +/- 0.000 (n=39)
condition=no_wall variant=standard_mppi_128_runtime	100	54/100 (54.0%)	0.134 +/- 0.045 (n=54)	-	260.098 +/- 4.521 (n=54)	37.981 +/- 1.536 (n=54)	0.003 +/- 0.000 (n=54)
condition=static_wall variant=control_bank_mppi_128	400	394/400 (98.5%)	0.281 +/- 0.068 (n=394)	0.281 +/- 0.068 (n=394)	259.550 +/- 26.457 (n=394)	50.282 +/- 5.591 (n=394)	0.005 +/- 0.001 (n=394)
condition=static_wall variant=corridor_prior_map	400	390/400 (97.5%)	0.201 +/- 0.054 (n=390)	0.201 +/- 0.054 (n=390)	257.941 +/- 18.162 (n=390)	47.795 +/- 6.752 (n=390)	0.004 +/- 0.001 (n=390)
condition=static_wall variant=frenet_corridor	400	395/400 (98.8%)	0.158 +/- 0.040 (n=395)	0.158 +/- 0.040 (n=395)	236.563 +/- 34.228 (n=395)	54.182 +/- 6.826 (n=395)	0.005 +/- 0.001 (n=395)
condition=static_wall variant=full_swarm_prior_map	400	398/400 (99.5%)	0.197 +/- 0.053 (n=398)	0.197 +/- 0.053 (n=398)	245.757 +/- 32.129 (n=398)	53.136 +/- 5.706 (n=398)	0.006 +/- 0.002 (n=398)
condition=static_wall variant=gaussian_prior_map	400	393/400 (98.2%)	0.150 +/- 0.037 (n=393)	0.150 +/- 0.037 (n=393)	233.285 +/- 34.705 (n=393)	54.934 +/- 7.345 (n=393)	0.006 +/- 0.001 (n=393)
condition=static_wall variant=mode_selecting_corridor	400	398/400 (99.5%)	0.189 +/- 0.046 (n=398)	0.189 +/- 0.046 (n=398)	262.168 +/- 17.562 (n=398)	46.492 +/- 5.888 (n=398)	0.010 +/- 0.004 (n=398)
condition=static_wall variant=mode_selecting_gaussian	400	397/400 (99.2%)	0.142 +/- 0.034 (n=397)	0.142 +/- 0.034 (n=397)	232.795 +/- 34.335 (n=397)	54.935 +/- 7.313 (n=397)	0.012 +/- 0.004 (n=397)
condition=static_wall variant=standard_mppi_128	400	60/400 (15.0%)	0.162 +/- 0.048 (n=60)	0.162 +/- 0.048 (n=60)	256.303 +/- 5.115 (n=60)	38.283 +/- 1.027 (n=60)	0.002 +/- 0.000 (n=60)
condition=static_wall variant=standard_mppi_128_runtime	400	87/400 (21.8%)	0.134 +/- 0.039 (n=87)	0.134 +/- 0.039 (n=87)	260.556 +/- 4.598 (n=87)	38.218 +/- 1.350 (n=87)	0.013 +/- 0.002 (n=87)

Failure reasons
Counts and percentages among failed trials

Case / variant	Failure reasons
condition=dynamic_wall variant=control_bank_mppi	not_reaching: 81 (100.0%)
condition=dynamic_wall variant=corridor_prior_mppi	collision: 1 (0.9%); not_reaching: 109 (99.1%)
condition=dynamic_wall variant=frenet_corridor_mppi	not_reaching: 115 (100.0%)
condition=dynamic_wall variant=full_swarm_prior_mppi	not_reaching: 87 (100.0%)
condition=dynamic_wall variant=gaussian_prior_mppi	not_reaching: 138 (100.0%)
condition=dynamic_wall variant=mode_selecting_corridor_mppi	not_reaching: 102 (100.0%)
condition=dynamic_wall variant=mode_selecting_gaussian_mppi	not_reaching: 127 (100.0%)
condition=dynamic_wall variant=standard_mppi	not_reaching: 346 (100.0%)
condition=dynamic_wall variant=standard_mppi_128_rollouts	not_reaching: 310 (100.0%)
condition=no_wall variant=control_bank_mppi	none
condition=no_wall variant=corridor_prior_mppi	not_reaching: 2 (100.0%)
condition=no_wall variant=frenet_corridor_mppi	not_reaching: 2 (100.0%)
condition=no_wall variant=full_swarm_prior_mppi	none
condition=no_wall variant=gaussian_prior_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=mode_selecting_corridor_mppi	none
condition=no_wall variant=mode_selecting_gaussian_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=standard_mppi	not_reaching: 61 (100.0%)
condition=no_wall variant=standard_mppi_128_rollouts	not_reaching: 46 (100.0%)
condition=static_wall variant=control_bank_mppi	not_reaching: 6 (100.0%)
condition=static_wall variant=corridor_prior_mppi	not_reaching: 10 (100.0%)
condition=static_wall variant=frenet_corridor_mppi	not_reaching: 5 (100.0%)
condition=static_wall variant=full_swarm_prior_mppi	not_reaching: 2 (100.0%)
condition=static_wall variant=gaussian_prior_mppi	not_reaching: 7 (100.0%)
condition=static_wall variant=mode_selecting_corridor_mppi	not_reaching: 2 (100.0%)
condition=static_wall variant=mode_selecting_gaussian_mppi	not_reaching: 3 (100.0%)
condition=static_wall variant=standard_mppi	not_reaching: 340 (100.0%)
condition=static_wall variant=standard_mppi_128_rollouts	collision: 1 (0.3%); not_reaching: 312 (99.7%)